

Lesson 3.1: The Carbon Cycle

Location: Classroom

Grade Level: High School

Time: One class session of 50 minutes

Driving Question

- How does carbon cycle through the spheres (hydrosphere, atmosphere, geosphere, and biosphere)?

Lesson Synopsis

During this 50-minute session, students begin exploring how carbon moves through Earth's systems, the hydrosphere, atmosphere, geosphere, and biosphere, and explore how human activities can influence this cycle. Students begin their reflections of the sources of carbon dioxide within their communities with the goal to understand how these sources influence global processes.

What students will figure out:

- How carbon moves through a system naturally.
- How human activities have altered the carbon cycle.
- The rate of change of carbon emissions impacts the oceans and other earth systems.
- Scientific models help us predict and understand these interactions and how changes to a system may impact other systems.

LEARNING OUTCOMES

By the end of this investigation, students will be able to...	You can assess this using...
1. Explain and describe the trend of the change of atmospheric carbon levels between 1750 and today.	Class discussion during the <i>Investigate, Navigate</i> section.
2. Draw a model that conveys how carbon naturally cycles through Earth's spheres and how human impacts have impacted this cycle.	Group models during the <i>Investigate</i> section.
3. Predict how a change in the rate of carbon emissions can impact Southern California's ocean and what their community values about the ocean.	Class discussion during the <i>Investigate</i> section.

Building towards NGSS and AI Literacy:

- **HS-ESS3-6:** Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity. In this lesson, students will construct a model based on local climate change and how human activity contributes to positive and negative effects in the environment.

Lesson Overview

Section	Time	What Happens
Investigate	10 minutes	Students analyze a slow-reveal graph showing how the average atmospheric level of carbon dioxide has changed over time, and then share their initial ideas about how Earth's climate is changing.
Investigate	30 minutes	Students begin to create a model of how carbon naturally cycles through Earth's spheres and then layer on how human sources of carbon emission have impacted this cycle and parts of the system.
Navigate	15 minutes	After sharing what they found, students take a moment to reflect on how carbon emissions have impacted the natural system. Using information from their community interviews and their models, they make a prediction on how this may impact what their community values about Southern California's ocean and choose a natural process (sea level rise, coastal erosion, weather patterns, ocean acidification and food webs) that they want to explore more about.
Problematize	5 minutes	Students take a moment to reflect on how their thinking has changed and identify what next steps they may need to take to learn more about their chosen processes to investigate.

MATERIALS

- [Lesson 3.1 Slideshow](#)
- [Science Journal](#)
- Concentrations vs. emissions from <https://www.climate.gov/media/12990>
- Whiteboards or large paper sheets (one per group) for models

BEFORE YOU START TEACHING

1. Include a numbered list of general steps to set up the investigation before students arrive.

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: In this lesson, students will use the information they have gathered from the previous module to create a model of the carbon cycle. The model will account for how human activities have altered the carbon cycle, and how the rate of change might impact the ocean and the earth systems.

Students might encounter and use the terms: **carbon cycle, carbon sinks, fossil fuels, combustion, decomposition, greenhouse gas, carbon footprint**; as well as terms related to the Earth's sphere, such as **atmosphere, biosphere, and hydrosphere**.

Where We Are NOT Going: At the end of the lesson, students identify a natural process (sea level rise, coastal erosion, weather patterns, ocean acidification, and food webs) connected to their model that they want to explore more about. We will investigate some of these processes in depth in later lessons in Module 3.

LEARNING SEQUENCE

Lesson plan notes are also provided on the slide notes.

INVESTIGATE: Analyze Changes of Carbon Dioxide Over Time (10 minutes)

1. Open the slideshow to **Slide 1** and inform students that today is the beginning of module 3. Tell students that in module 3, they will focus on the carbon cycle and incorporate it into the models they have built in Modules 1 and 2.
2. Advance to **Slide 2** and have students work in groups to think about their models from Module 2.
 - How did you collect data to inform your SageModeler model?
 - What information is still missing? How can you get that information?

After students have discussed, share with the class that in Module 3, they will explore how they can strengthen their models even further through additional investigations.

3. Advance to **Slide 3** and share the objectives for lesson 3.1. Tell them that in Module 3, they will be focusing further on carbon dioxide's role in their model and how it can affect other factors.
4. Move to **Slide 4**, which shows a dual axis line graph without any labels.

The light blue line, which correlates to the y-axis on the left, shows increases in global atmospheric carbon dioxide levels since 1751. The gray line, which correlates to the y-axis on the right, shows increases in carbon dioxide emissions since 1751.

Invite students to make observations about the graph:

- What do you notice? What do you wonder?
 - How would you describe the trends that you see?
5. Advance through **Slide 5**, which will add numerical labels to the graph's x-axis. As you do so, ask students:
 - What do you notice? What do you wonder?
 6. Move on to **Slide 6**, which will show the labeled left-hand y-axis. Ask the class:
 - What do you notice? What do you wonder?
 - What do you think is causing the trend we see in the graph?

Once students share, confirm for them that the light blue line is a visual representation of the average amount of carbon dioxide in our atmosphere.

7. Have student groups discuss together and answer the following question. When they have discussed, bring the class back together.
 - What do you think might have caused this rise in atmospheric carbon dioxide levels?
8. Advance to **Slides 7**, which adds labels to the gray y-axis on the right, revealing that it shows an increase in carbon dioxide emissions. Ask students:
 - What do you notice? What do you wonder?
 - How are these graphs similar? Different?
9. Finally, move on to **Slide 8**, which gives the graph a title and shares the original source of the graph. Once you've finished revealing the full graph, ask the class:
 - How would you describe the relationship between these graphs?
 - Where might the carbon dioxide emissions be coming from?
 - How do you think the rise in carbon dioxide in our atmosphere might be affecting our planet?
10. As students share, reinforce ideas that discuss the increase in carbon dioxide emissions over the years or how it can impact their community. Use the opportunity to assess their current understanding, not to correct misconceptions.
11. After students have shared, tell students that today, they will start to learn more about carbon dioxide and how it cycles throughout our planet.

INVESTIGATE: Create a Model of the Carbon Cycle (30 minutes)

1. As you move into the Investigate section, tell students that in order to understand how carbon is moving through a system, you are going to have them work in their groups to build a drawn model of how carbon naturally moves and interacts with the earth.
2. Advance to **Slide 9**, where they will see a base model and a few things to keep in mind while modeling. [Science Journal, Module 3.1 The Carbon Cycle, Student Task: Create a Model of the Carbon Cycle \(p. 25-26\)](#) has the same reflection questions for students to think about as they build their models.

As you model, think about how carbon interacts with different parts of the Earth, including the:

- Geosphere: The solid parts of Earth, including rocks, soil, and fossils.
- Biosphere: All the living things on Earth, including plants, animals, and humans.

- Hydrosphere: All the water on Earth, including oceans, ice, and moisture in the air.
- Atmosphere: The air we breathe and the gases surrounding Earth.

Also, make sure you are including:

- Where carbon is found in your model
 - Labels to describe the pieces of your drawing
 - Arrows to show the movement of carbon between spheres
3. Hand out a poster paper for students to build their model as a group. As students are working on their model, walk between groups and use guiding and probing questions to help them think about how to expand on their model.
 - How is carbon moving between the atmosphere, hydrosphere, biosphere, or geosphere in this section?
 - Which processes are involved here? (photosynthesis, respiration, decomposition?)
 - How might carbon move into the ocean or other parts of the hydrosphere?
 - Which processes return carbon to the atmosphere?
 - Where might carbon go next?
 4. As students are finishing up, bring them together to have them think about how carbon might be introduced through human sources. Move to **Slide 10** and invite students to consider how they may add human sources of carbon into their model.
 - Which processes in your model are human contributions?
 - What other human sources of carbon can you think of?

As students share, listen for ideas focused on deforestation, fossil fuel burning (electricity and heat production; transportation), agriculture, industry emissions (e.g., from cement production), and commercial and residential buildings.

5. Once students have shared, give them a few minutes to update their model to include these new ideas. To help them make connections, ask:
 - Which parts of the natural cycle are most affected by these activities? Where does the extra carbon go?

NAVIGATE: Reflect on How Carbon Emissions Impact the Natural System (15 minutes)

1. Bring the class back together and advance the slideshow to **Slide 11**. Give a moment for students to share their findings as a class. Ask the following question to get the conversation started.
 - How did carbon move through the Earth in the model?
 - How might rising CO₂ emissions through human activities impact the carbon cycle?
2. Once students have shared, ask them to think back to what their community values about the Southern California ocean. Advance to **Slide 12** and ask them to share responses as a class to the following questions:

- Which aspects of the ocean do you or your community care about?
- How could changes in the carbon cycle or increases in carbon emissions affect those values?
- How might we explore further?

As students share their ideas, listen for and recognize responses that focus on exploring processes we will focus on in Module 3, which include causes of sea level rise, coastal erosion, changing global temperatures, changes in the ocean's pH, and impacts on food webs.

Notes to educators: Depending on which focus students show more interest in (e.g., sea level rise/coastal erosion versus ocean acidification/food webs), select the appropriate lesson, EITHER 3.3 OR 3.4. Students may do both if you have time.

PROBLEMATIZE: Identify Next Steps (5 minutes)

1. As you wrap up for the day, share with students that their final task for the day is to reflect on the lesson.
2. Advance to **Slide 13** and have student teams work together to answer the following questions in their Science Journals on [Module 3.1: The Carbon Cycle \(Student Reflection\)](#), p. 27.
 - How has your thinking about the carbon cycle changed? What new ideas or connections did you discover about how carbon moves through the Earth's systems and how humans affect that movement?
 - What natural process did you choose to investigate next (e.g., sea level rising, coastal erosion, ocean acidification, or weather pattern changes? How does this process affect the value your community places on Southern California's ocean?
 - What questions do you still have? What evidence, data, or models would help you understand this process better?
3. As students wrap up their discussion, and if time allows, have teams share with the entire class.
4. Thank the students for their time today. Let students know that during the next session, they are going to take more time to explore how carbon dioxide can impact the Earth's atmosphere, and how that impacts humans.